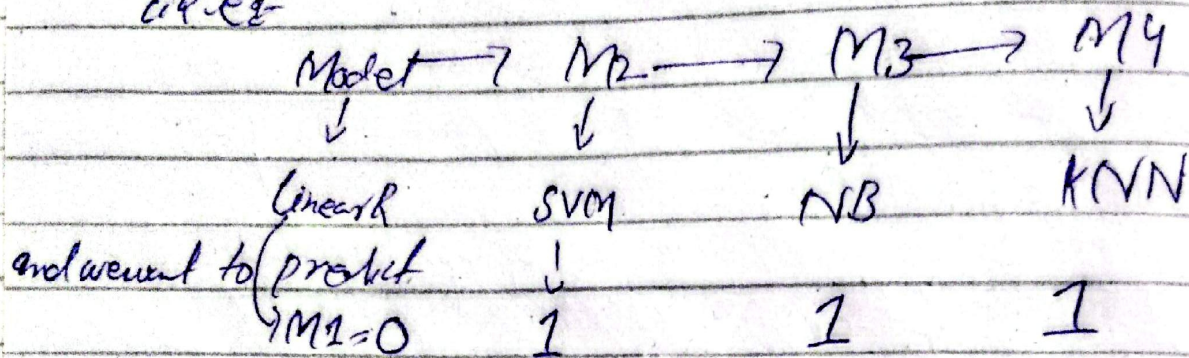


⇒ Note: The word Ensemble (means to listen Majority).
Ensemble Learning!
ex: we hear the crowd! on the basis of it
ensemble ^{searching} way created

in ML, we have data → train/test → Models
we create Multiple Models in Ensemble Learning
like eg



so 3 Models says true means that
particular thing should happen and one says
no.
so obviously we will choose Majority

There are some types of Ensemble Learning
we will discuss later.
so this was the core concept of
ensemble Learning.

⇒ Note: we can also work on Ensemble Learning
on Regression.

just what you have to do is
just find mean instead of voting.

Types of Ensemble Learning.
There are mostly 3 types.

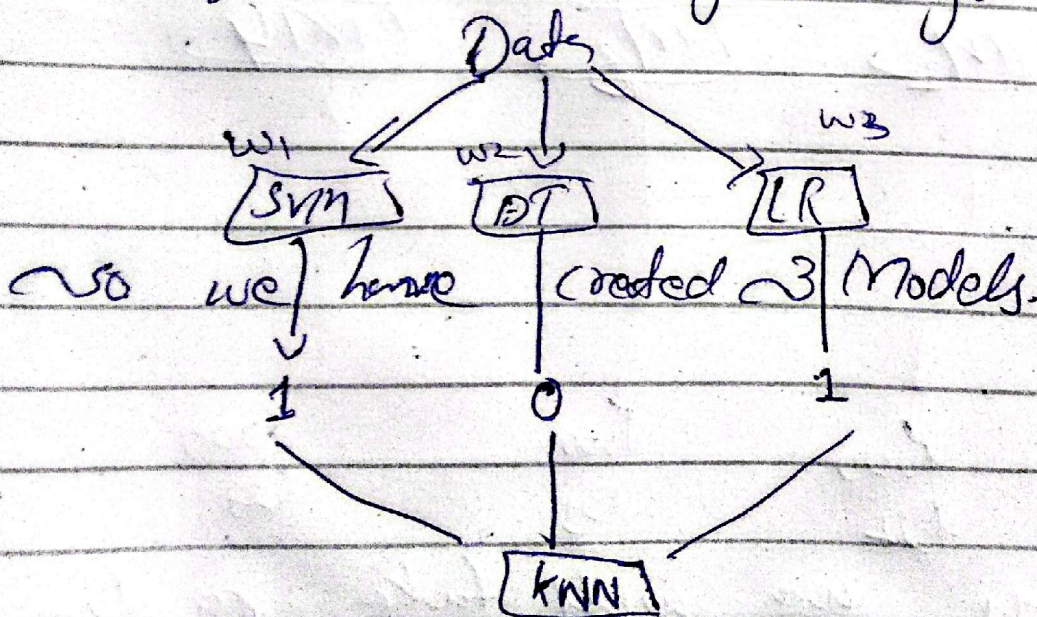
1. Bagging
↓
Randomforest

2. Boosting

→ AdaBoost
→ Gradient Boost
→ XGBoost
(extreme boost)

3. stacking.

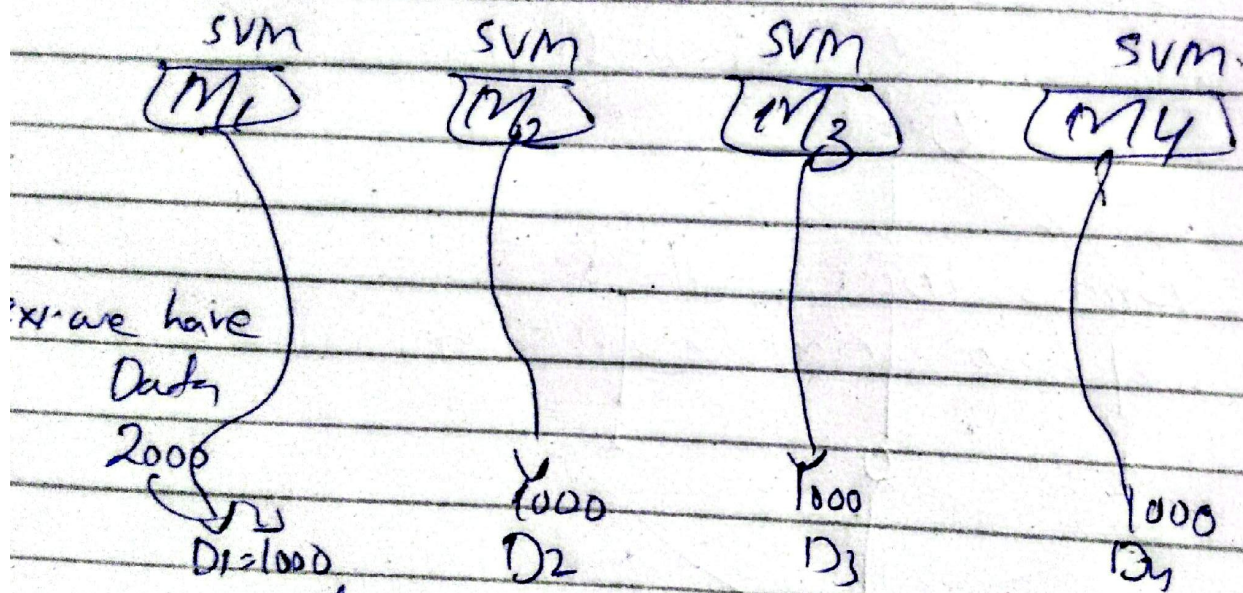
Stacking: stacking the multiple models performance. we create multiple models from one Data and this is useful for classification and Regression Alg.



and the final prediction will come on the basis of their weights

⇒ Bagging! it is much effective than
Stacking.
its full form is bootstrap Aggregation.

in bootstrap we take random samples
so we create models. but in all
models we will use one ~~Algorithm~~ Algorithm only.
ex.

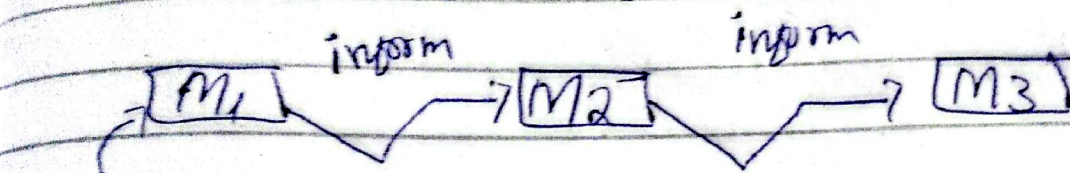


always we will take random
data points from each.

so for each model will learn again and
again. and then we will see
the result which comes more, we
use again Voting technique.

Q3

Boosting! is the one of the best technique which used



Data
2000

step 1: all data will train to M_1 and the mistakes ~~he~~ make will inform to M_2 and M_2 will inform to M_3 and so on.